ELECENG 3EY4
Sample Problems on Coordinate Frames and Homogenous Transformations

1- Consider the following chain of rotations

$$R_4^0 = R_y(\theta_1)R_z(\theta_2)R_y(\theta_3)R_x(\theta_4).$$

Express various ways of interpreting the rotations, about fixed and current axes that would yield the same result.

2- Find the ZYZ Euler representation for the following rotation matrix

$$R = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ -1 & 0 & 0 \end{bmatrix}$$

3- Consider the figure below. A robot is set up 1 meter from a table, two of whose legs are on the y axis as shown. The table top is 1 meter high and 1 meter square. A frame $o_1 - x_1y_1z_1$ is fixed to the edge of the table as shown. A cube measuring 20 cm on a side is placed in the center of the table with frame $o_2 - x_2y_2z_2$ established at the center of the cube as shown. A camera is situated directly above the center of the block 2 m above the tabletop with frame $o_3 - x_3y_3z_3$*o* attached as shown. Find the homogeneous transformations relating each of these frames to the base frame $o_0 - x_0y_0z_0$. Find the homogeneous transformation relating the camera to the object frame.

Now suppose that, after the camera is calibrated, it is rotated $90°$ about the axis *z*. Re-compute the above coordinate transformations.

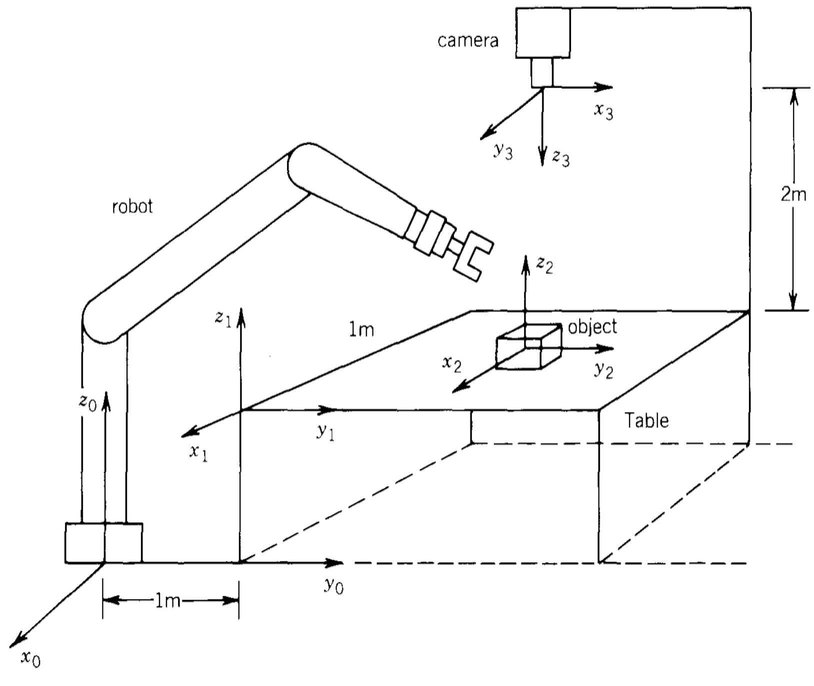


Fig. 1 (adopted from M. Spong and M. Vidyasagar, Robot Dynamics and Control, John Wiley & Sons)

.

4- In general, multiplication of homogeneous transformation matrices is not commutative. Consider the matrix product:

$$T = \mathrm{Rot}_{x,\alpha}\mathrm{Tran}_{x,b}\mathrm{Tran}_{z,d}\mathrm{Rot}_{z,\theta}$$

Here Rot and Tran are purely rotational and translational homogenous transformations. Determine which pairs of the four matrices on the right-hand side commute. Explain why the pairs commute. Find all permutations of these four matrices that would yield the same homogenous transformation matrix, *T*.

5- Consider the wedge-shaped object in Fig. 2. In this figure, Frame A and Frame B are attached to the rigid object whereas Frame 0 is a stationary reference frame. (Note that frames 0 & A initially coincide)

a. Find the homogenous transformation T_B^0 that relates Frame B to Frame 0 (0 is origin and B is destination).

b. The coordinates of point **P** in Frame B are given by $P^B = \begin{bmatrix} 0 & 0 & 2 \end{bmatrix}^T$. Find the coordinates of this point in Frame 0, P^0.

c. Assume that the wedge-shaped object is moved first by translating Frame A along its x-axis by -1 units and then rotating this frame about its current y-axis by $\frac{\pi}{2}$ radians. Calculate the new coordinates of point P in Frame B, $\overline{P}^B$. (Note that point P is *not* attached to the rigid object and therefore has *not* moved itself).

d. Assume that the movement of the object is restricted to pivoting around the origin of Frame A, i.e., the origin of the frame *cannot* move. Find a rotation for Frame A that would bring the origin of Frame B to its closest possible distance from point P. Does this problem have a unique solution? Explain.

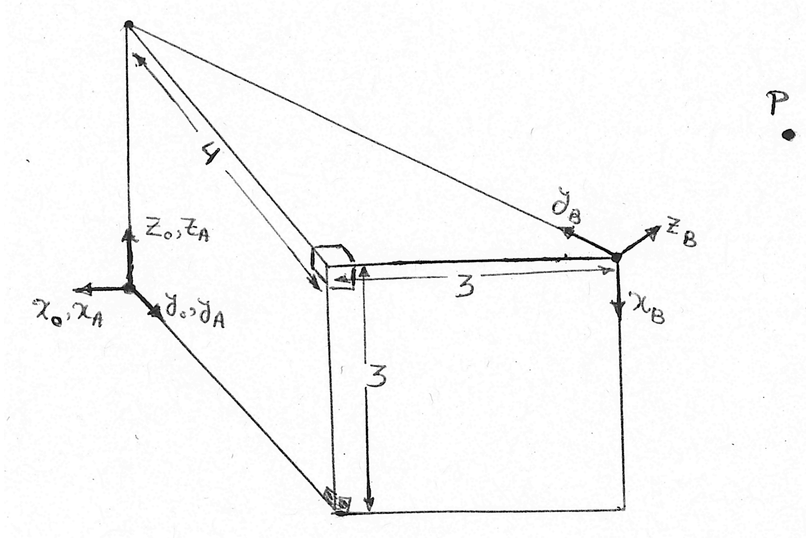


Fig. 2. The wedge-shaped object in Q5.

[redacted]

1)

- Rotations about current axes (Left to right)
- Rotations about fixed axes (right to left)

- $\xrightarrow[\text{current } y]{\theta_3} R_y(\theta_3) \xrightarrow[\text{current } x]{\theta_4} R_y(\theta_3)R_x(\theta_4) \xrightarrow[\text{fixed } z]{\theta_2} R_z(\theta_2)R_y(\theta_3)R_x(\theta_4)$
  $\xrightarrow[\text{fixed } y]{\theta_1} R_4^0$

- $\xrightarrow[\text{current } y]{\theta_3} R_y(\theta_3) \xrightarrow[\text{fixed } z]{\theta_2} R_z(\theta_2)R_y(\theta_3) \xrightarrow[\text{fixed } y]{\theta_1} R_y(\theta_1)R_z(\theta_2)R_y(\theta_3)$
  $\xrightarrow[\text{current } x]{\theta_4} R_4^0$

other combinations can similarly be obtained.

2)

(i) $0 < \vartheta < \pi \quad (s\vartheta > 0)$

$$\varphi = \operatorname{atan2}(r_{23}, r_{13}) = -3\pi/4$$
$$\psi = \operatorname{atan2}(r_{32}, -r_{31}) = 0$$
$$\vartheta = \operatorname{atan2}\left(\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right) = \pi/2$$

(ii) $-\pi < \vartheta < 0 \quad (s\vartheta < 0)$

$$\varphi = \operatorname{atan2}(-r_{23}, -r_{13}) = \pi/4$$
$$\psi = \operatorname{atan2}(-r_{32}, r_{31}) = \pi$$
$$\vartheta = \operatorname{atan2}\left(-\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right) = -\pi/2$$

Spatial Description of a Rigid Body

ZYZ Euler Angles $\Phi = \begin{bmatrix} \varphi & \vartheta & \psi \end{bmatrix}^T$:

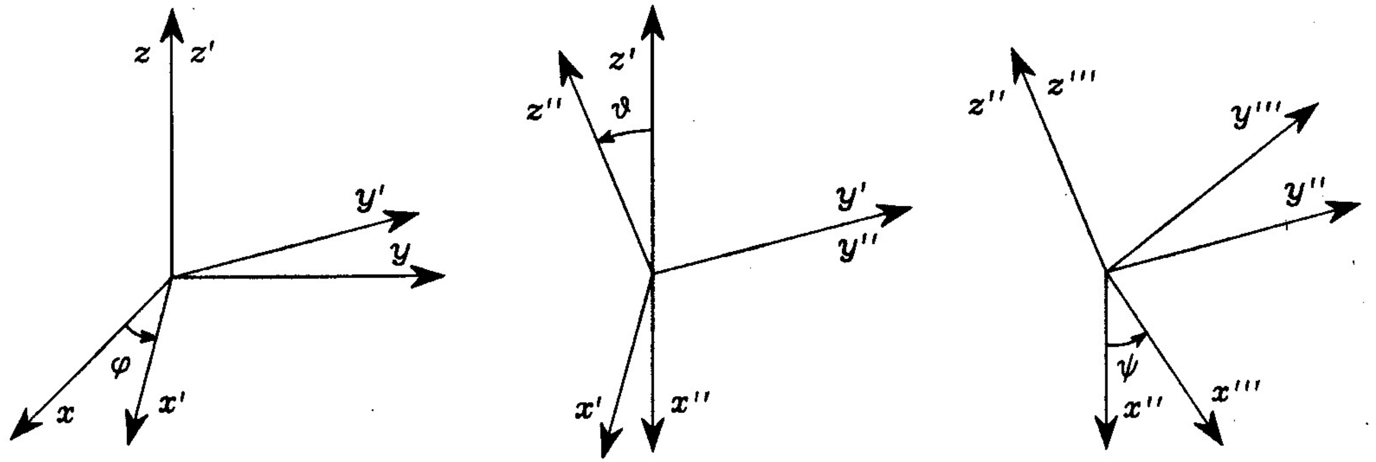
$R_z(\varphi)$: Rotation about z axis by the angle φ

$R_{y'}(\vartheta)$: Rotation about the current y' axis by the angle ϑ

$R_{z''}(\psi)$: Rotation about the current z'' axis by the angle ψ

Equivalent rotation matrix

$$\boldsymbol{R}(\phi) = \boldsymbol{R}_z(\varphi)\boldsymbol{R}_{y'}(\vartheta)\boldsymbol{R}_{z''}(\psi) = \begin{bmatrix} c_\varphi c_\vartheta c_\psi - s_\varphi s_\psi & -c_\varphi c_\vartheta s_\psi - s_\varphi c_\psi & c_\varphi s_\vartheta \\ s_\varphi c_\vartheta c_\psi + c_\varphi s_\psi & -s_\varphi c_\vartheta s_\psi + c_\varphi c_\psi & s_\varphi s_\vartheta \\ -s_\vartheta c_\psi & s_\vartheta s_\psi & c_\vartheta \end{bmatrix}$$



Spatial Description of a Rigid Body

ZYZ Euler Angles $\Phi = \begin{bmatrix} \varphi & \vartheta & \psi \end{bmatrix}^T$

Reverse procedure: Given the rotation matrix

$$R(\Phi) = \begin{bmatrix} c_\varphi c_\vartheta c_\psi - s_\varphi s_\psi & -c_\varphi c_\vartheta s_\psi - s_\varphi c_\psi & c_\varphi s_\vartheta \\ s_\varphi c_\vartheta c_\psi + c_\varphi s_\psi & -s_\varphi c_\vartheta s_\psi + c_\varphi c_\psi & s_\varphi s_\vartheta \\ -s_\vartheta c_\psi & s_\vartheta s_\psi & c_\vartheta \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

What are the possible sets of ZYZ Euler angles?

$$\varphi = \text{Atan2}(r_{23}, r_{13})$$
$$\vartheta = \text{Atan2}\left(\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right)$$
$$\psi = \text{Atan2}(r_{32}, -r_{31}).$$

$0 < \vartheta < 180$

$$\varphi = \text{Atan2}(-r_{23}, -r_{13})$$
$$\vartheta = \text{Atan2}\left(-\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right)$$
$$\psi = \text{Atan2}(-r_{32}, r_{31}).$$

$-180 < \vartheta < 0$

Special case: when $\vartheta = 0 \text{ or } 180 \Rightarrow \sin(\vartheta) = 0$ and $r_{13}=r_{23}=r_{31}=r_{32}=0$. In this case, we have two consecutive rotations about two parallel axes and only $\varphi+\psi$ or $\varphi-\psi$ can be found.

[redacted]

3) [redacted]

$$A_1^0 = \left[\begin{array}{c|c} I & \begin{matrix}0\\1\\1\end{matrix} \\ \hline 0^T & 1 \end{array}\right] \qquad A_2^0 = \left[\begin{array}{c|c} I & \begin{matrix}-0.5\\1.5\\1.1\end{matrix} \\ \hline 0^T & 1 \end{array}\right]$$

$$A_3^0 = \left[\begin{array}{ccc|c} 0 & 1 & 0 & -0.5 \\ 1 & 0 & 0 & 1.5 \\ 0 & 0 & -1 & 3 \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \qquad A_2^3 = \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1.9 \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$$

After camera rotation! (about z_3)

$$A_3^0 = \left[\begin{array}{ccc|c} 1 & 0 & 0 & -0.5 \\ 0 & -1 & 0 & 1.5 \\ 0 & 0 & -1 & 3 \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \qquad A_2^3 = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 1.9 \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$$

4) [redacted]

a- $$T_B^A = T_B^0 = \left[\begin{array}{ccc|c} 0 & 3/5 & & -3 \\ 0 & -4/5 & z_B & 4 \\ -1 & 0 & & 3 \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \qquad z_B = x_B \times y_B$$

b- $$\left[\begin{array}{c} p^0 \\ \hline 1 \end{array}\right] = T_B^0 \left[\begin{array}{c} p^B \\ \hline 1 \end{array}\right]$$

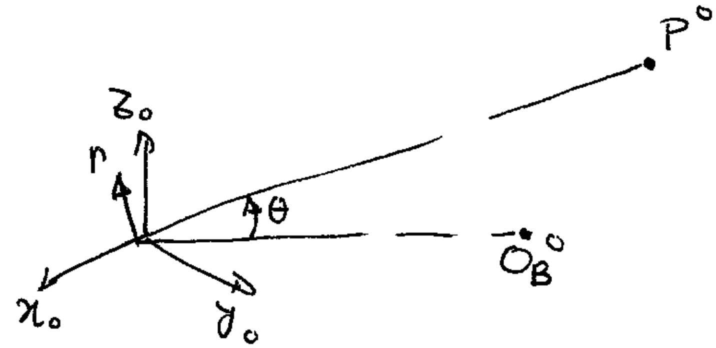
from part (a)

c- $$T_{B'}^0 = \left[\begin{array}{c|c} R_y(\pi/2) & \begin{matrix}-1\\0\\0\end{matrix} \\ \hline 0 & \end{array}\right] T_B^A$$

From part (b)

$$\begin{bmatrix} \bar{P}_B \\ \hline 1 \end{bmatrix} = T_0^{B'} \begin{bmatrix} P^0 \\ \hline 1 \end{bmatrix}$$

d -



$$r = \frac{O_B^0 \times P^0}{\|O_B^0 \times P^0\|} \qquad \theta = \cos^{-1}\left(\frac{P^0 \cdot O_B^0}{\|P^0\| \|O_B^0\|} \right)$$

$$R_A^0 = R\left(\underbrace{v}_{\text{any } v \text{ would}}, \frac{P^0}{\|P^0\|}\right) R(\theta, r)$$

any $\underline{\underline{v}}$ would $\Rightarrow$ an <u>infinite</u> number of solutions exist.

5) [redacted]

$$T = Tran_{x,b}\, Rot_{x,\alpha}\, Tran_{z,d}\, Rot_{z,\theta}$$

$$T = Tran_{x,b}\, Rot_{x,\alpha}\, Rot_{z,\theta}\, Tran_{z,d}$$

$$T = \overset{Rot_{x,\alpha}}{\cancel{Tran_{x,b}}}\, \overset{Tran_{x,b}}{\cancel{Rot_{x,\alpha}}}\, Rot_{z,\theta}\, Tran_{z,d}$$

$$T = Rot_{x,\alpha}\, Tran_{z,d}\, Tran_{x,b}\, Rot_{z,\theta}$$